Bagian VI Daftar Pustaka dan Abstrak

Deteksi Gegenpressing Menggunakan Temporal Graph Neural Network pada Data Tracking IDSSE

Abstrak dan pustaka tersedia, tetapi angka abstrak berasal dari pilot 264 label. Verifikasi syarat tahun dan peringkat SINTA sebelum pengumpulan.

Draf kerja • hasil pilot 264 label; anotasi penuh 1.873 episode belum selesai.

Abstrak

Penelitian ini mengembangkan dan mengevaluasi pipeline untuk mendeteksi gegenpressing setelah kehilangan penguasaan bola menggunakan data posisi dan event IDSSE. Sebanyak 2.353 kandidat transisi diperiksa; 1.873 episode memenuhi kriteria valid ketat. Dari 264 episode yang dipilih untuk anotasi, seluruhnya telah memperoleh label biner: 222 gegenpressing dan 42 bukan gegenpressing. Episode direpresentasikan sebagai urutan graph selama empat detik, dari dua detik sebelum sampai dua detik sesudah kehilangan bola, pada 25 Hz. Model Temporal GNN yang menggabungkan dua lapisan Graph Convolutional Network dan GRU dibandingkan dengan logistic regression serta XGBoost melalui pembagian data berdasarkan pertandingan. Pada 40 sampel uji dari satu pertandingan, Temporal GNN memperoleh F1 0,870 dan recall 0,938, sedangkan XGBoost memperoleh average precision tertinggi, yaitu 0,906. Logistic regression menghasilkan precision tertinggi, yaitu 0,839. Hasil menunjukkan pertukaran kinerja antarmodel, bukan keunggulan yang konsisten dari satu model. Karena test hanya mencakup satu pertandingan dan delapan contoh negatif, temuan ini dibatasi sebagai proof of concept dan belum dapat digeneralisasikan ke pertandingan profesional secara luas.

Kata kunci: gegenpressing; counterpressing; tracking sepak bola; graph neural network; IDSSE

Abstract

This study develops and evaluates a pipeline for detecting gegenpressing after possession loss using IDSSE positional and event data. Of 2,353 transition candidates, 1,873 episodes met strict structural validity criteria. All 264 selected episodes received binary annotations: 222 positive and 42 negative. Each episode was represented as a four-second graph sequence spanning two seconds before and after possession loss at 25 Hz. A Temporal GNN combining two Graph Convolutional Network layers with a GRU was compared with logistic regression and XGBoost using match-level splits. On 40 test episodes from one match, the Temporal GNN achieved an F1 score of 0.870 and recall of 0.938, while XGBoost achieved the highest average precision at 0.906. Logistic regression had the highest precision at 0.839. The results show metric trade-offs rather than consistent superiority of one model. Because the test set contains a single match and only eight negative examples, findings should be treated as a proof of concept and cannot yet be generalized to professional football more broadly.

Keywords: gegenpressing; counterpressing; soccer tracking; graph neural network; IDSSE

Daftar Pustaka

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